Akhilaa Sonduri Panthangi

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Education

University of Southern California — Masters of Science in Computer Science **Aug 2023 – May 2025**
Coursework: Analysis of Algorithms, Web Technologies, Database Systems, Machine Learning, Natural Language Processing, Information Retrieval

Technical Skills

- **Languages & Core**: C,C++,Python, JavaScript, TypeScript, SQL, Swift, HTML/CSS.
- **AI / ML & Backend**: LLMs, RAG, FAISS, Neo4j, NLP, Prompt Engineering, Agentic AI, Multi-Agent Systems, Embeddings, FastAPI, Node.js, REST APIs, Microservices.
- **Frontend, Databases & Cloud**: React.js, Next.js, .NET | MongoDB, PostgreSQL, Redis | AWS, Azure, Docker, CI/CD, Git.
- **Certifications**: **Claude Certified Architect – Foundations (CCA-F)** , AWS Certified Cloud Practitioner.

Experience

Proliant Data **Remote**
AI Engineer *May 2026 – Present*

- Delivered the core **Python** library behind **RESOLVE**, an **AI-native, NLP-to-SQL** agent that answers analysts' questions in natural language over enterprise migration data, built as ∼3,300 lines across 12 modules (safety layer, metadata registry, **Neo4j** KG queries, **FAISS** retrieval) grounding every generated SQL query and answer in **hybrid RAG**.
- Cut token spend and eliminated runaway loops in a **28-tool autonomous agent** through **prompt caching**, history pruning, and result compaction, backed by hard termination guards (budget caps, forced synthesis) the model cannot override.
- Secured the platform to enterprise standards, closing **SQL-injection, CORS, MIME-spoofing**, and credential risks via **AST-based SQL validation**, **Azure Key Vault** rotation, and **PII redaction**, gated by a **CI eval harness** on correctness, groundedness, and cost.

University of Southern California - Viterbi School of Engineering **Los Angeles, CA**
Research Assistant *Sep 2025 – Apr 2026*

- Architected **agentic AI frameworks** and **multi-agent systems** using **React, TypeScript, Node.js, Express.js, and MongoDB**, enabling real-time visualization of **LLM reasoning** and optimizing **information extraction, ranking, and adaptive planning** workflows.
- Built **MoodMap**, an AI-powered location recommendation engine using **Groq Llama 3.3 70B**, **RAG**, and **OpenStreetMap** for mood-based personalized suggestions via a mobile-first chatbot interface [Try it here]; integrated **LLMs, NLP, and RAG** pipelines with responsive frontend dashboards and APIs, enhancing coordination accuracy, reducing latency, and improving scalability across complex research prototypes.

HubSpot **Cambridge, MA**
Software Engineer *Jun 2025 – Sep 2025*

- Built an **AI-powered summary and suggestion panel** in the **Workflow Action Builder**, parsing triggers, filter branches, and action steps into structured payloads, reducing **pre-activation review time by 30%** and **workflow misconfiguration by ∼20%**.
- Enhanced **workflow efficiency by 25%** by designing a new input panel to configure, reorder, validate, and save multiple workflow inputs with improved usability and visual consistency.
- Led a **data model migration** for workflow categories, defining clear data structures and service contracts to enable **scalable automation** and consistent platform behavior.
- Boosted **platform reliability and QA efficiency by 30%** by integrating real-time APIs, strengthening feature controls, execution safeguards, and monitoring for billing and analytics.

WorkUp **Los Angeles, CA**
Software Developer Intern *May 2024 – Jul 2024*

- Engineered an **AI-Driven Interview Customization Feature** and **Automated Template Selection** system in **React and Node.js**, increasing **user adoption by 40%** and cutting recruiter setup time by **50%**.
- Revamped the **AI Copilot Web App UI** by migrating to **React** and embedding **video playback** of candidate responses in reports, improving **decision-making by 60%** and overall user engagement by **30%**.

Accolite Digital **Hyderabad, India**
Software Developer *Jan 2022 – Jul 2023*

- Administered **AWS services** (**VPCs, RDS, S3, EC2, Lightsail**) and orchestrated **CI/CD pipelines** with containerized applications as a **Certified AWS Cloud Practitioner**, slashing **deployment time by 20%** while ensuring secure, scalable infrastructure; led migration to **GCP**, cutting **operational costs by 15%**.
- Optimized **Git workflows** and collaborated cross-functionally on **Node.js, .NET, React, microservices, and REST API integration**, strengthening monitoring and alerting across cloud infrastructure and increasing system **reliability and uptime by 25%**.

Projects

NutriCraft: Food, Macro and Cost Tracker GitHub | *React Native, SQLite, Drizzle ORM, Supabase, PostgreSQL*

- Engineered an offline first React Native app with a local SQLite database via Drizzle ORM that syncs to Supabase Postgres, so users can log meals with no connection and stay consistent across devices.
- Added a tool calling chat and voice assistant that converts free text meal descriptions into structured food, macro, and cost records, cutting manual entry to a single message.

Real-time Stock Search Web App Try it here | React, Node.js, MongoDB, SwiftUI, RWD

- Developed a responsive web and iOS application using **React, Node.js, MongoDB Atlas, Bootstrap, and Swift**, providing real-time stock data retrieval and portfolio management.
- Integrated **Finnhub API and Polygon AI** for live stock insights and managed user portfolios with MongoDB, increasing engagement and user satisfaction by **75%**.

Publications

Cryptocurrency Price Prediction using ARIMA | Python, Machine Learning, Time Series

- Predicted **Ethereum daily price changes** with **95% accuracy** leveraging **ARIMA** and advanced **time series analysis**.
- Disseminated **research findings** in **IEEE Xplore**, contributing to advancements in **cryptocurrency forecasting**.